MRT pilot: a design and power note

Oct 31, 2026 · @Niko

This pilot did not test its hypothesis. It was designed to estimate whether new MRT and light-rail stations in Taiwan raised new business registrations within 500 m. Before any outcome data was opened, the planned analysis was run on synthetic data with known answers. The amended inference (D2) failed that check twice, in the D2 gate and the D5 gate. Under the pre-agreed rules (D6), no outcome file is opened for H1.

This note reports what was learned instead:

- **Two measured results** from the simulations: the kill rules jointly cap the chance of a claim at about 75%, even when the effect is real; and the registered small-cell rule biases the estimate down by 3.9% of the effect.
- **Two explanations that remain unproven**: a conjecture about the shared base quarter (Finding 1), and a probable mechanism for the D5 failure (Finding 2).
- The simulation record, failures included.
- The design's power: the primary test alone reaches 80% power at an effect of about +20%. Under the full decision rule ($p < 0.05$ and no kill rule), 80% power is never reached.

All numbers come from synthetic data or plan documents, with one disclosed exception, D1 (section 2). A short glossary is in section 8.

1. The hypothesis, the plan and its amendments

H1. In the 12 quarters after opening, new business registrations per 1,000 residents in 村里 within 500 m of a new MRT or light-rail station rise relative to comparison 村里. The lines are 機場捷運 (with A22), 淡海輕軌 (綠山線 and 藍海線), 環狀線 and 安坑輕軌, opened 2017–2023. **H0** is no difference.

The registered method was a staggered difference-in-differences (Callaway and Sant'Anna 2021) with Ferman–Pinto inference, kill rules and 27 robustness tests.

Table 1. The plan and its recorded amendments.

Record	What it says	Commit (main, nikolasdoan/tw-econ-research)
Pre-registration v2.1	The binding plan, signed by Niko and a reviewer	a743f88 (merged 2026-10-31T01:34:00Z)
D1	Three whole-window export totals were posted before the merge; no 村里, quarter or station figure	b43cdcd
D1a	Corrects D1 on who saw the totals, and when: the research team received them at 01:51 UTC, after the merge (section 2)	b5bcdaf
D2	Referee ruling on analysis code v1: placebo-district inference, revised K2b, R28, r*, and a simulation gate	255eb68
D3	機捷 opening-date source (data.gov.tw 96918 row, 2017-03-02); k = 0 unchanged	6c9c970
D4	Reserved for Appendix A (see section 5)	not yet committed
D5	Referee ruling on the failed D2 gate: K2a and K2b at 0.05, a line × quarter variance term, one rerun on fresh seeds	6c9c970
D6	Referee ruling: D5 applies as written; the gate fails; no outcome file opens for H1; this note is written instead	e1e151c

Labelling. The referee's own ruling on code v1 called its inference fix "D1" and its gate "D2". In `deviations.md` that whole ruling is committed as D2, because D1 was already the export disclosure. The referee's D5 and D6 are committed under the same numbers. This note uses the `deviations.md` numbers throughout.

Code v1 is at f56e15b; code v2 and the D2 gate results are at ba06c4e. Code v3 and the D5 gate results are at 8962b74. D3 and D5 share commit 6c9c970 because they were committed together.

Data sources and licences. The plan admits only open-licence sources. The data are described by source and licence only.

Table 2. Data sources named in the plan.

Data	Source	Licence, as the source page states it
New registrations and closures	Registry records linked by 統一編號. Publisher: 財政部財政資訊中心 (Ministry of Finance), 全國營業(稅籍)登記資料集, data.gov.tw 9400	政府資料開放授權條款-第1版 (Open Government Data License v1.0)
村里 population	內政部戶政司 (data.gov.tw 8411, 32973)	政府資料開放授權條款 v1
村里 boundaries	內政部地政司 村里界歷史圖資, version 114年 (data.gov.tw 130549)	政府資料開放授權條款 v1
Housing prices (secondary outcome)	內政部 不動產交易實價登錄. Page opened: 「本期發布之不動產買賣實價登錄批次資料」, data.gov.tw 25119, agency listed as 地政司	政府資料開放授權條款-第1版
Stations and opening dates	交通部鐵道局, 「鐵道局轄管計畫已完成通車啟用時間」, data.gov.tw 96918; plus the dated operator and news pages listed in the pre-registration, cited for dates only	Dataset: 政府資料開放授權條款-第1版. Nothing from the cited pages is reused.

The three data.gov.tw pages for 9400, 25119 and 96918 were opened on 2026-10-31. Each page's 授權方式 field names 政府資料開放授權條款-第1版. Page 9400 also links the licence text at data.gov.tw/license. Page 25119 is the periodic batch release; the full price series used in the export is recorded in its manifest.

2. What real data was seen, by whom and when

This section states the one exception to "no outcome data" plainly.

- **The export.** The data export was built on the project's M1 machine by the data builder. No registration, closure, price or population-series file from it was opened, requested or read by the research team, its agents or the referee.
- **The cloud manager.** Before the plan was merged (2026-10-31T01:34:00Z), the data builder posted three whole-window totals: the number of sales in the price file, the number of registrations for 2011Q1–2026Q3, and the number of suppressed cells per file. The project's coordinating agent (the cloud manager) saw them. They are recorded as D1 (b43cdcd). None of them is broken down by 村里, quarter, station, line, district or ring.
- **The research team.** D1's committed text says the research team did not see the totals. That was true at the merge, but not afterwards. At 01:51 UTC the cloud manager sent the research team the totals so that it could write D1's wording. From then on, the research team had seen the three totals. D1a (b5bcdaf) records this correction.
- **The referee** saw none of them, as far as the research team knows.

Did real data shape the design? No. The synthetic base rate of 1.2 registrations per 1,000 residents per quarter was written into the synthetic-data code at 01:27 UTC. That was before the totals reached the research team (01:51) and before the merge (01:34). It is a round assumption by the code's author. It was not taken from any data source or publication, and it has no citation. No design value was changed after 01:51. The D2 and D5 code inherited the same 1.2.

With that exception stated, every other number in this note comes from synthetic data made by our own code, or from plan documents. H1 is therefore **untested**. This note is not evidence for or against it.

3. Four simulation rounds

Each seed builds a fresh synthetic city: 5,000 村里 on a 40 × 40 km plane, with the real lines' opening quarters and a known true effect τ in registrations per 1,000 residents per quarter. The assumed base rate is 1.2 (section 2), so τ = 0.5 is about +42%. Across seeds, the city has 67–129 treated 村里 in total (median 96–99 by round), and the smallest line has 38–39 comparison districts.

In Table 3, rates are shares of seeds, with Monte Carlo (MC) SEs in percentage points in brackets. **Bold** marks a value outside the limit in force for that round. Round 1 had no limits, so nothing in it is bold. "—" means not computed. Every figure in rounds 2–4 is recomputed from the per-seed files by `note_tables.py` (Attached files, end of section 3).

Table 3. Simulation results by round.

Round	Scenario (seeds)	Bias (SE)	Rejection at 5%	95% coverage	K2a	K2b	K3a	K3b	Any kill
1. v2.1 as registered (calibration check, no limits)	Effect, τ = 0.5 (5)	+0.027	—	Spread ±0.1 vs half-width ±0.07	3 of 5	2 of 5	—	—	4 of 5
2. D2 gate	Null (500)	+0.002 (0.004)	5.6% (1.0)	94.2% (1.0)	**14.0% (1.6)**	**25.0% (1.9)**	8.4% (1.2)	5.0% (1.0)	41.2% (2.2)
	Effect (200)	−0.019 (0.006)	power 100%	**88.0% (2.3)**	**15.0% (2.5)**	**21.5% (2.9)**	7.0% (1.8)	3.5% (1.3)	41.0% (3.5)
	Pre-trend (200)	—	—	—	65.5% (3.4)	86.5% (2.4)	6.0% (1.7)	16.0% (2.6)	98.0% (1.0)
	Stress (200)	−0.006 (0.008)	**8.0% (1.9)**	**92.0% (1.9)**	12.0% (2.3)	**26.5% (3.1)**	1.5% (0.9)	4.5% (1.5)	39.0% (3.4)
3. D2 diagnostics (same seeds, reported only)	Effect, no suppression (200)	0.000 (0.006)	power 100%	93.0% (1.8)	—	—	—	—	—
	Round-2 runs re-scored with K2a and K2b at 0.05	—	—	—	5.0–8.0%	5.5–7.4%	—	—	16.5–23.8%; pre-trend 92.0% (1.9)
4. D5 gate (fresh seeds, σ^2_j on)	Null (500)	+0.004 (0.004)	7.0% (1.1)	93.0% (1.1)	8.2% (1.2)	7.2% (1.2)	8.0% (1.2)	4.2% (0.9)	24.0% (1.9)
	Effect, suppressed (200)	−0.027 (0.006)	power 100%	91.5% (2.0)	7.5% (1.9)	7.5% (1.9)	9.0% (2.0)	4.5% (1.5)	25.0% (3.1)

Round	Scenario (seeds)	Bias (SE)	Rejection at 5%	95% coverage	K2a	K2b	K3a	K3b	Any kill
	Effect, no suppression (200)	−0.007 (0.006)	power 100%	**91.5% (2.0)**	8.5% (2.0)	6.5% (1.7)	8.5% (2.0)	5.0% (1.5)	24.0% (3.0)
	Pre-trend (200)	—	—	—	47.5% (3.5)	72.0% (3.2)	6.0% (1.7)	17.5% (2.7)	89.0% (2.2)
	Stress (200)	−0.009 (0.007)	2.0% (1.0)	98.0% (1.0)	4.0% (1.4)	7.0% (1.8)	2.5% (1.1)	1.5% (0.9)	13.5% (2.4)

Limits. Round 2 (D2): rejection 3.0–7.5%, coverage ⩾ 92.5%, each kill rule ⩽ 12.5%, pre-trend any kill ⩾ 80%. Round 4 (D5) used the same limits, plus two for the effect scenario: bias ⩽ 0 and |bias| ⩽ 10% of τ under suppression, and coverage ⩾ 92.5% **without** suppression. Round 4 failed only on that last limit. With σ^2_j off, round 4′s null also falls outside: rejection 7.6% (1.2) and coverage 92.4% (1.2).

Which inference each round used. Round 1 used v2.1′s registered inference. Rounds 2–4 used the amended placebo-district inference of D2, with D5′s changes in round 4. So the inference that failed the two gates was the amended one, not the registered one. Each round is reported as run, and none was rerun to change its outcome.

What can be checked. Round 1′s per-seed outputs were not kept, except for one seed (20261031). Its figures come from the run log and cannot be recomputed. That is one reason Finding 1 is rated a conjecture. Every figure in rounds 2–4 and in the power table can be recomputed from the attached files.

Attached files (uploaded to this note; SHA-256 in full):

- `note_tables.csv`: every rate, MC SE, bias, half-width ratio and layout count used in this note. Asset fcf1273e3c8a45d88176ff69e28caf4f, SHA-256 5908a1a56c4e8b5e068b05b0494c34fab2c102a4b19d84287330a3dab85a0b43.
- `note_tables.py`: the script that builds it from the per-seed files. Uploaded as `note_tables.py.txt`: asset 815f234319cb2c8338229005e77e1a8a, SHA-256 d4c99de790296506aa3018f10dfb68d7e9482cdacdb392784ec4c15d9319a787.
- `note_d2_power_seeds.csv`: per-seed results for rounds 2–3 and for the power runs at τ = 0.1–0.4 (2,100 rows). Asset f2eab441136f0a2636a1d661a8020a27, SHA-256 0f9376828b2ed457a54fac8a4683658b8bc749aecde25e9a2fd8bbd097b2da95.
- `power_summary.csv`: Asset 9bd1c269f466b89b6ea839cb4f5d0276, SHA-256 45577eebd6f225c5b615a27a00cc2d4873d8b30f1b326f4a151209e4e3b7a975.

- `gate_v3_all_seeds.csv` (round 4, per seed): attached to the "analysis code v3 and D5 gate" doc, SHA-256 61e28dafdab4d174b5e9daac971b83839ee4882f5f048c6f7a0d4f16c85b483b.

4. Findings about the design

There are four results, in two kinds. Results A and B are measured directly in the simulations. Findings 1 and 2 are explanations. Neither has been proven, and each is rated as such.

Result A: the kill rules cap the chance of a claim at about 75%. With a real effect of τ = 0.3–0.5 and no pre-trend, the four kill rules each fire 4.5–9% of the time. Together they fire in 22–27% of seeds. So even a large true effect is claimed ($p < 0.05$ and no kill) in only 73.0–76.0% of seeds (MC SE about 3 points; Table 4). Any successor design should budget its kill rules jointly.

Result B: the small-cell rule biases the estimate down. The registered rule imputes suppressed counts of 1–2 as 1.5. On the D5 effect seeds ($\tau = 0.5$), the estimate's bias is −0.0266 (SE 0.0058) with suppression, which is 5.3% of τ. Without suppression it is −0.0072 (0.0061). On the same seeds, suppression itself moves the estimate by −0.0194 (paired SE 0.0016), which is 3.9% of τ. A plausible reason is that the treatment moves counts out of the 1–2 band, which changes what the imputation hides. The bias works against H1.

Finding 1 (conjecture): one shared base quarter may make standard few-cluster inference too confident. With a universal base quarter, every treated 村里 in a cohort is measured against the same single quarter ($k = -13$). Noise in that quarter moves every event-time estimate of that cohort together. Comparison districts enter many cells with different base quarters, so their noise averages out. Ferman–Pinto builds the null from comparison-district residuals, so it would understate the treated side's noise.

The evidence is weak:

- In round 1 the estimate varied by about ±0.1 across 5 seeds, while each interval was about ±0.07 wide. K2a fired in 3 of the 5.
- Those 5 seeds cannot be recomputed (section 3).
- No test varied the base period, or compared a cohort-shared shock with and without the fix.

- The remedy that followed (D2) gives each comparison district a placebo contribution built exactly like a treated line's: the same cohort dates, base quarter, cells and weights. With it, null rejection was 5.6–7.0% and null coverage 93–94% (rounds 2 and 4). But D2 changed several things at once, so this does not isolate the base quarter as the cause.

We chose to rate it a conjecture rather than run a new check during this revision. A targeted check, with a varying base period or a cohort-shared shock run with and without the fix, belongs to any successor study.

Finding 2 (probable mechanism): the treatment may add noise that untreated placebos cannot see. Registrations are counts. When the treatment raises the treated rate by about 40%, it also raises the treated counts' Poisson variance. Placebo residuals come from untreated districts, so they would miss that extra variance. The evidence, with its uncertainty (σ^2_j on, D5 seeds, no suppression for the effect):

- The ratio of interval half-width to the estimate's SD is 1.90 (MC SE 0.06) under no effect and 1.71 (0.08) under $\tau = 0.5$. The gap of 0.20 is about 2.0 combined MC SEs.
- Coverage is 93.0% (1.1) under no effect and 91.5% (2.0) under $\tau = 0.5$. The gap of 1.5 points is under one combined SE (2.3).
- The test's size under no effect is 7.0% (1.1).

The pattern fits the mechanism. But the coverage gap is within MC noise, and the D5 coverage miss (91.5% against a 92.5% limit) is itself within about one MC SE of the limit. This note therefore does not claim that this mechanism caused the D5 failure.

5. Appendix A, power and the minimum detectable effect

Appendix A (D4). No pre-outcome design facts exist yet: the pre-outcome bundle (`data/pre`) has not arrived, so Appendix A is not written and nothing about the real layout is reported here.

Power, from synthetic data only. The synthetic layout is described in section 3. Real treated counts and comparison districts may differ, so these are design numbers, not predictions.

Table 4. Power and claim rate by true effect (MC SEs in percentage points in brackets).

True effect (per 1,000 residents per quarter)	Approx. % of the assumed base	Seeds and source	Power of the primary test at 5%	95% coverage	Claim rate: p < 0.05 **and** no kill rule fires
0	0%	500; D5 gate null seeds 50001–50500	7.0% (1.1)	93.0% (1.1)	2.2% (0.7)
0.1	+8%	200; power seeds 91001–91200	22.5% (3.0)	94.5% (1.6)	18.0% (2.7)
0.2	+17%	200; power seeds 91001–91200	69.0% (3.3)	94.5% (1.6)	57.5% (3.5)
0.3	+25%	200; power seeds 91001–91200	94.5% (1.6)	94.5% (1.6)	75.5% (3.0)
0.4	+33%	200; power seeds 91001–91200	100%	92.0% (1.9)	73.0% (3.1)
0.5	+42%	200; D5 gate effect seeds 60001–60200	100%	91.5% (2.0)	75.0% (3.1)

Every row uses the registered small-cell imputation (suppressed 1–2 counts set to 1.5) and σ^2_j on. The $\tau = 0$ and $\tau = 0.5$ rows are the D5 gate's own runs, so the $\tau = 0.5$ row is the gate's "effect, suppressed" scenario. The $\tau = 0.1$–0.4 rows are new runs on seeds 91001–91200 (not committed in advance; this is a power curve, not a gate). The source of each row is in `power_summary.csv` and `note_tables.csv` (section 3).

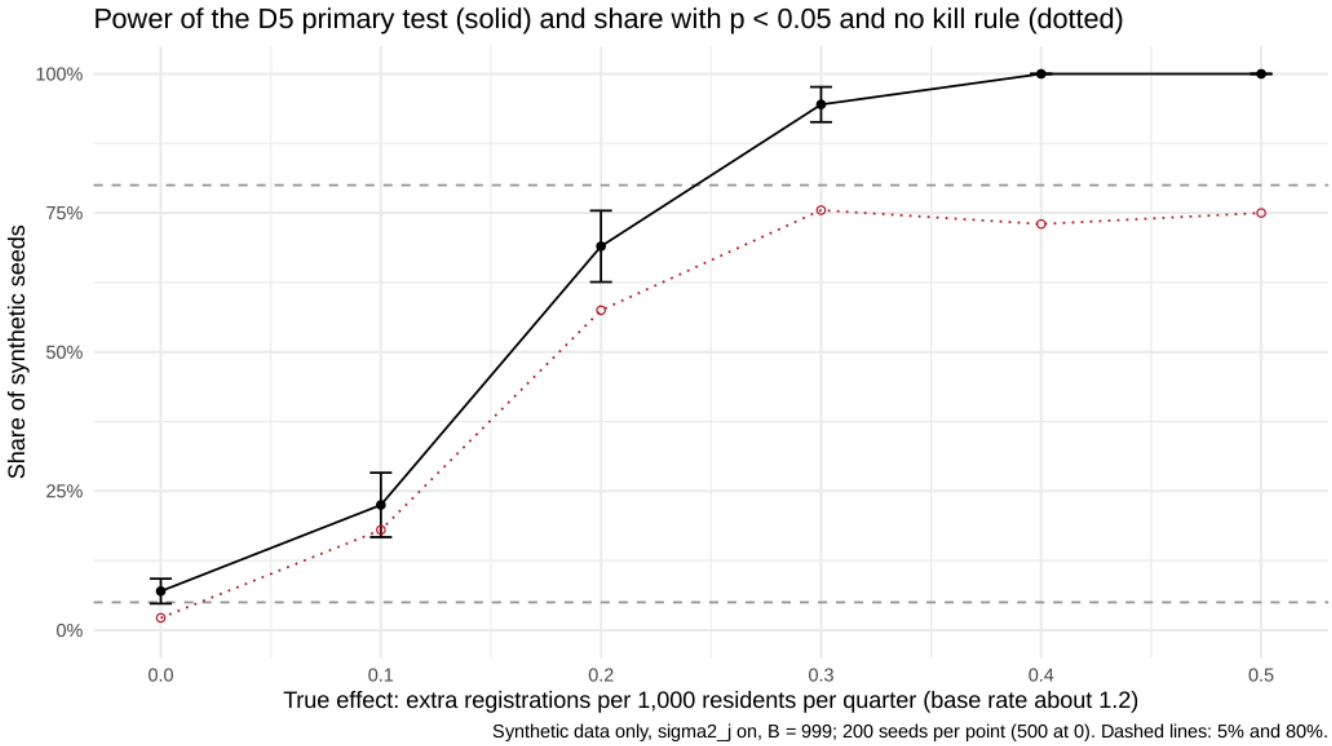


Figure 1. Power of the D5 primary test (solid) and claim rate, p < 0.05 with no kill rule (dotted), by true effect. Synthetic data only, from the same runs as Table 4. Bars are 95% MC intervals for the primary test's power. The dashed lines mark 5% and 80%.

Minimum detectable effect (MDE). The two definitions give different answers.

- **Primary test alone.** It reaches 80% power at about **0.24 registrations per 1,000 residents per quarter**, interpolating between 0.2 (69.0%) and 0.3 (94.5%). That is about +20% on the assumed base of 1.2. The estimate's SD is about 0.08 at every effect size.
- **Decision rule (p < 0.05 and no kill rule).** In these runs, 80% power is never reached. The highest claim rate is 75.5% (MC SE 3.0) at τ = 0.3, and the rate stays at 73–76% up to τ = 0.5 (Result A). So the decision-rule MDE does not exist for this design.

Code: `sim_power.R` (SHA-256 d3c7673f82bf03fba719dee5abb44ea7caad0b81d7420990d2c438d78e6cfb57) is `sim_gate_v3.R` with τ read from the scenario name.

6. Limits of this note

- **Everything rests on one synthetic world.** Counts are Poisson. Village and district effects are independent. Shocks are district-wide or line-wide, at 10%. The base rate of 1.2 is assumed (section 2). Real data may have spatial correlation, overdispersion or

trends this world lacks. The gate results hold for this world only.

- **Power is for a synthetic layout.** Real treated counts and N_0 come from Appendix A, which is not written. A smaller treated count or fewer comparison districts would raise the MDE.
- **The monitoring limits are coarse.** With 200 seeds, a coverage rate carries a Monte Carlo SE of about 2 points. Several results sit within one SE of their limits, on both sides.
- **Not independently replicated.** One agent wrote all the estimation and simulation code. The red team (stage 5) has not run. The engine matches `did` 2.5.1 to 11–12 digits where both apply, but the placebo inference and σ²_j have no outside check.
- **Approximations.** σ²_j is spread across event times in proportion to each line's weights. That is exact for the K2a mean and approximate for the K2b slope. The variance floor and the NNLS fit are our implementation of D2′s wording.
- **Not run.** HonestDiD (R15) was never run: it needs R ⩾ 4.4, and D6 ruled it not needed once H1 closed. No real-data robustness test was run, by design. The synthetic data-generating process is assumed, not estimated. The gate's own MC imprecision (SE about 2 points at 200 seeds) means results near a limit are partly luck.
- **Our own errors, found and fixed before any result was reported.** Each was caught by a check, not by looking at results:
  - a data.table scoping bug that gave every cohort all treated 村里 (estimate 0.31 instead of 0.54);
  - a locale fault that silently merged 5 treated clusters into 2;
  - an exploding variance fit (draw SD 121);
  - an indexing bug in the σ²_j weight function.

7. The referee reports

An independent AI referee reviewed the plan and each gate. It worked apart from the authors, and its rulings reached us verbatim through the coordinating agent. It estimated its spend at about US$6, US$14, US$18 and US$20 across its reads.

In Table 5, a verdict in quotation marks is the referee's own wording. The "Main points" column is our one-line summary. The rulings themselves are quoted after the table and committed at the listed commits.

Table 5. Referee rounds.

Round	Verdict	Main points (our summary)
Pre-registration v1	"FIX THEN SIGN. The design can answer its question. But 5 must-fixes leave the estimate open to choices made after the data are seen, or make the primary test unable to reject."	5 must-fixes: one opening-date rule; the 綠線 cutoff and identified cells; Ferman–Pinto cannot reach p < 0.05 with 3–5 control groups; the denominator fallback; data choices left to the export. Plus 8 should-fixes.
Pre-registration v2	Fix then sign (as relayed; the full report did not reach the authors)	Two problems the revision caused: a false claim that 4 lines cannot reach p < 0.05; a noise-prone slope condition in K2b.
v2.1	Sign (as relayed)	—
D2 (after our v1 simulation)	"APPROVE DEVIATION D1 BELOW. No outcome file may be opened until the simulation gate (D2) passes and the results are committed." (Its "D1" is the inference fix; see Labelling, section 1.)	Placebo-district inference, revised K2b, R28, r*, and a simulation gate
D5 (after the D2 gate failed)	"AMEND AND RERUN ONCE ON FRESH SEEDS."	K2a and K2b at 0.05; line variance σ^2_j; fresh seeds
D6 (after the D5 gate failed)	"D5 APPLIES AS WRITTEN. The gate fails. No outcome file is opened for H1."	This note instead

Errors in the referee's work, stated plainly:

1. **Round-1 report, R1 contributions.** It said 綠山線 keeps k ⩾ 5 when 2020–22 is dropped; by the quarters it keeps k = 0–3. The referee later agreed, and lists it among its errors.
2. **K2b's false fires.** In its ruling on code v1 the referee wrote: "K2b's false fires are partly MY doing: in N2 I removed its slope-significance condition." Its replacement fired K2b only when three conditions all held: the extrapolated pre-trend has the same sign as the estimate; it is at least 50% of the estimate's size; and its one-sided placebo tail is below 20%. In D5 the referee wrote: "my 20% tail fires at least 20% of the time by construction. That was my error." That overstates it. Because of the other two

conditions, a correctly sized rule could fire less than 20% of the time. The real problem is different: **the placebo tail itself was mis-sized.** In the D2 gate the tail alone fell below 20% in 25.0% (1.9) of null seeds, 23.0% of effect seeds and 26.5% of stress seeds, against a nominal 20%. The two extra conditions almost never bound under no effect, so K2b fired at 21.5–26.5%. Under D5 the tail fell below 5% in 7.2% (SE 1.2) of null seeds, against a nominal 5%, so the mis-sizing may persist at the 5% level.

3. **The gate's seed count.** In D6: "My own error, for the record: a limit of 92.5% tested on 200 seeds (SE about 2 points) gave a gate whose result is partly luck near the boundary. That flaw sits in my design of the gate, not in the authors' work. It does not change the ruling." Our arithmetic agrees. With 200 seeds, a method whose true coverage is 95% fails a 92.5% limit 4.4% of the time; at 94% it fails 14.9% of the time; at 93%, 32.7%.

The referee also caught real problems that we had missed. The most important were that v2's inference could never reach $p < 0.05$ without a larger comparison pool, and that the opening-date rule was applied inconsistently across lines.

Key passages from the rulings.

- Ruling on code v1 (committed as D2): "DIAGNOSIS: RIGHT, AND IT IS THE PLAN, NOT THE CODE. v2.1 §6 step 1 defines a comparison district's contribution as its average over every post cell it enters. Those cells span up to 6 cohorts, so 6 different base quarters. A treated line's contribution reuses one base quarter, $k = -13$, in all 12 cells." It rejected calibrating thresholds by simulation: "that would tune the kill rules to an assumed noise model; simulation is used only to validate."
- D5: "AMEND AND RERUN ONCE ON FRESH SEEDS. … If any hard limit fails on the fresh seeds, the gate cannot pass on a design that answers H1; the study is then written up as a short note on design and power, and no outcome file is opened for H1."
- D6: "D5 APPLIES AS WRITTEN. The gate fails. No outcome file is opened for H1. … A limit fixed in advance does not bend because the miss is small. Each of the three remedies would be a second rerun after seeing a failure, which is exactly what D5 ruled out." On Finding 2, D6 describes the treated-side noise as "making intervals about 10% narrow under a 40% effect". Section 4 rates that as a probable mechanism, not a proven cause.
- D6 on stage 3: no outcome file is opened "even 'descriptively'; any look would contaminate a successor study on the same data." HonestDiD and R 4.5 are not needed.

8. Glossary

- **k, and k = −13.** Event time in quarters relative to a line's opening quarter (k = 0). The plan allows 12 quarters of anticipation (k = −12 to −1). So every estimate is measured against one base quarter, k = −13, the last quarter before anticipation can start.
- **K1.** A kill rule on sample size. The study stops if fewer than 30 treated 村里 remain after exclusions.
- **K2a.** A kill rule on the pre-trend. It fires if the average pre-period effect (k = −20 to −14) differs from zero: $p < 0.10$ under D2, and $p < 0.05$ under D5.
- **K2b.** A kill rule on the pre-trend slope. Fit a straight line through the pre-period effects and extend it to k = +5.5. The rule fires if all three hold: the line has the same sign as the main estimate; it reaches at least 50% of the estimate; and its one-sided placebo tail is below 20% (D2) or 5% (D5).
- **K3a.** A placebo kill rule. It gives the planned-line (綠線) 村里 a fake opening in 2020Q1. It fires if the fake effect has the main estimate's sign, has $p < 0.10$, and is at least a third of the main estimate's size.
- **K3b.** A placebo kill rule. It moves each treated line's opening 16 quarters earlier and uses only real pre-period data. It fires under the same three conditions as K3a, against the k = 0–3 estimate.
- **r* (break-even ratio).** The treated 村里's post-opening level divided by the level implied by their base quarter plus the comparison change. Under the plan, an effect with $r^* < 1.25$ is not reported as real in absolute terms.
- **R15.** The robustness test using HonestDiD (Rambachan–Roth bounds on violations of parallel trends). Not run (section 6).
- **R28.** A robustness test that replaces the single base quarter with the mean of k = −16 to −13 (added by D2; reported only).
- **NNLS.** Non-negative least squares. It is used to fit the placebo variance as $a + b/n$, where n is the number of 村里, with a and $b \geq 0$ and a floor of 0.25 × the mean squared residual.
- **σ^2_j.** The variance of line-wide, quarter-specific shocks for line j. It is estimated from the 800–1,500 m ring (R2) over k = −20 to −1 and added to the placebo draws (D5).
- **MC SE.** Monte Carlo standard error: the uncertainty of a simulated rate due to the finite number of seeds, $\sqrt{(p(1-p)/n)}$.